

MUHAMMAD TALHA

AI/ML Engineer | Computer Vision & Deep Learning Specialist

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SUMMARY

AI/ML Engineer with expertise in Computer Vision, Deep Learning, and Full-Stack Development. Proven track record of building production-grade ML applications using PyTorch, TensorFlow, and modern web frameworks. Currently pursuing Bachelor's in Computer Science at COMSATS University with hands-on experience from Software Productivity Strategists internship and CapregSoft bootcamp. Specialized in developing intelligent systems leveraging CNNs, NLP, and real-time image processing for impactful solutions.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C++

AI/ML Frameworks: PyTorch, TensorFlow, Keras, Hugging Face Transformers, scikit-learn

Computer Vision: OpenCV, MediaPipe, CNNs, Image Processing, Real-time Detection

Web Development: React.js, Node.js, Express.js, FastAPI, MongoDB, REST APIs

DevOps & Tools: Docker, Kubernetes, Git, GitHub, JWT Authentication, Postman

Data Science: Pandas, NumPy, Matplotlib, Altair, Data Visualization, NLP

PROFESSIONAL EXPERIENCE

Freelancer | Upwork

February 2025 - Present

- Deliver AI/ML and web development solutions to global clients, specializing in computer vision and NLP applications

Software Engineering Intern | Software Productivity Strategists, Inc. (SPS)

July 2025 - August 2025

- Contributed to dynamic web development projects using modern frameworks and tools in professional environment
- Collaborated with cross-functional teams to deliver scalable solutions following industry best practices

Software Engineering Bootcamp | CapregSoft (Private) Limited

March 2025 - April 2025

- Mastered full-stack web development with React.js, Node.js, Docker, and Kubernetes through intensive hands-on training
- Gained expertise in API development, JWT authentication, and deployed production-ready applications on GitHub Pages

KEY PROJECTS

SignStream Pro - Real-Time Sign Language Detector (SaaS) | Jan 2026 - Feb 2026

- Architected full-stack ASL recognition web app with React, TypeScript, Tailwind CSS frontend and FastAPI backend
- Implemented real-time hand landmark extraction using OpenCV and MediaPipe for precise gesture detection
- Deployed TensorFlow/Keras deep learning model achieving high accuracy in A-Z sign language classification

Campus Issue Tracker (MERN Stack) | Dec 2025 - Jan 2026

- Developed comprehensive issue tracking platform with React/Vite frontend featuring interactive Recharts dashboards

- Built secure Node.js/Express/MongoDB backend with JWT authentication, Multer file uploads, and Nodemailer integration
- Implemented PDF export functionality and automated email alert system for streamlined university issue management

Empathy AI - Sentiment Analysis Chatbot / *Sep 2025 - Nov 2025*

- Built intelligent chatbot using PyTorch and Hugging Face Transformers (RoBERTa, BERT) for real-time sentiment analysis
- Engineered Streamlit application with live emotional probability metrics dashboard using Pandas and Altair
- Deployed context-aware NLP system generating empathetic responses based on continuous emotion tracking

Digit Recognizer with GUI / *Jun 2025 - Jul 2025*

- Created desktop application using Tkinter and TensorFlow for real-time handwritten digit recognition (0-9)
- Trained CNN on MNIST dataset achieving 98%+ accuracy with PIL/NumPy preprocessing and Matplotlib visualization

EDUCATION

Bachelor of Science in Computer Science

COMSATS University Islamabad, Wah Campus | February 2023 - February 2027 (Expected)

- Focus: Artificial Intelligence, Machine Learning, Computer Vision, Deep Learning, Software Engineering

CERTIFICATIONS

- AI For Beginners
- Python for Data Science, AI & Development
- Python for Data Analysis: Pandas & NumPy